

SIMATSENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 10%

1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.
2. A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.
3. An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.
4. A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.
5. An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.
6. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.
7. An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.
8. An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.
9. Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

10. A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.
11. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.
12. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.
13. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.
14. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.
15. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.
16. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.
17. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.
18. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.
19. Detected edges in an image appear broken. Explain and justify a method to improve edge continuity, including its role in post-processing.
20. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.
21. An image suffers from excessive brightness variation between regions. Explain and justify a suitable histogram processing technique to normalize intensity distribution, describing its working and expected outcome.
22. A medical image requires enhancement of subtle intensity differences. Describe and justify a suitable grey level transformation to highlight fine variations, explaining its effect on diagnostic visibility.
23. An image contains salt-and-pepper noise. Explain and justify the most appropriate spatial filter, describing its working mechanism and effectiveness in noise removal.
24. An image appears overly smooth after filtering, losing important features. Explain and justify a sharpening technique to restore details, including its working and impact on edges.

25. An image contains both low-frequency background and high-frequency noise. Describe and justify a filtering approach that preserves important structures while reducing noise, explaining the concept clearly.
26. An image requires enhancement of edges without amplifying noise significantly. Explain and justify a suitable spatial or frequency domain method, describing its operation and expected result.
27. A grayscale image shows poor separation between object and background intensities. Describe and justify a thresholding approach that improves segmentation accuracy, explaining its working.
28. An image contains multiple regions with similar intensity but different textures. Explain and justify a region-based segmentation approach, describing how it groups pixels effectively.
29. Detected edges are noisy and fragmented. Describe and justify a method to refine edges and improve boundary detection, explaining its role in segmentation.
30. An application requires precise object boundary extraction for shape analysis. Explain and justify a suitable segmentation and boundary detection approach, describing its working and effectiveness.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>

CO2-AT3 - Class Test

1. A dark image lacks details. Describe and justify appropriate grey level transformation to enhance visibility, explaining the improve of low-intensity.

Power-law Transformation:

Transformation increases the brightness of dark pixels while preserving the natural appearance.

Effects:

- Reveals hidden details
- Preserves the natural contrast of brighter areas.
- Produces a clearer and readable image.

Justification:

Specifically enhances low-intensity pixels without increasing brightness.

Applications:

- Medical image enhancement
- Night-time photography
- CCTV Surveillance

Conclusion:

Power-law Transformation is the most suitable for dark images. It effectively enhances low-intensity regions.

2. A noisy image contains random intensity variations. Explain the suitable smoothing filter.

Gaussian Filter:

Reduces random intensity variations. It smooths the image by giving higher weight to central pixel and gradually lower weights to neighboring pixels.

Effects:

- Smooths the image uniformly
- Preserves overall image structure
- Reduces random intensity noise.

Advantages:

- Reduces random noise.
- Produces smooth and natural image
- Preserves image features.

Conclusion:

Effectively reduces noise using weighted average.

3. An image appears blurred after smoothing. Explain suitable sharpening techniques.

Laplacian filter:

Laplacian filter is a suitable sharpening method because it enhances regions where there is a rapid change in intensity, such as edges and fine details.

- Enhances edges.
- Restores fine details
- Improves image sharpness.

Expected Results:

- Restores details lost during smoothing
- Improves edge visibility.
- Enhances image clarity.

The Laplacian Filter is preferred as it enhances edges by emphasizing high-frequency information

4. An image contains high-frequency noise. Explain the appropriate frequency domain filter.

Low-Pass Filter (LPF):

Allows low-frequency components to pass while suppressing high-frequency noise.

Effects:

- Reduces high-frequency noise
- Produces a smoother image
- Preserves the overall image structure.

The LPF is preferred because high-frequency noise mainly exists in high-frequency region of frequency spectrum. By suppressing these components, the filter removes noise while retaining the important image content.

- Digital photography
- Remote sensing.

5. A smooth version of an image is required.
Describe the frequency domain technique.

Low-Pass Filter (LPF):

LPF is suitable frequency domain technique. It allows low-frequency components to pass while attenuating high-frequency components.

→ Removes high-frequency noise

→ Produces smooth image.

→ Reduces unwanted intensity variations.

After applying LPF:

→ Noise is significantly reduced.

→ Image becomes smoother.

6. An object is clearly brighter than the background.
Explain the segmentation technique used.

Global Thresholding:

Global Thresholding is used for object and background have a clear difference in intensity.

Working Process:

Pixel Intensity $> T \rightarrow$ Object

Pixel Intensity $\leq T \rightarrow$ Background

T - Threshold value

Effectiveness:

→ Separates objects from the background.

→ Produces a binary image.

→ Improves object detection.

→ Simplifies further analysis.

→ Simple and fast.

7. Separate foreground and background. Explain Segmentation method.

Thresholding:

→ Pixels with intensity greater than T are classified as Foreground.

→ Pixels with intensity less than T are classified as background.

Justification:

Threshold is simple, fast and highly effective.

Applications:

→ Object detection

→ Industrial inspection

→ CV Systems.

8. An image contains low-frequency background and high-frequency noise.

Low-Pass Filter (LPF):

→ Convert image using Fast Fourier Transform.

→ Spatial domain using Inverse Fast Fourier.

Effects:

→ Removes high-frequency noise.

→ Preserves the image structure.

→ Produces smoother image.

→ Improves image quality.

9. An image requires enhancement of edges without amplifying noise. Explain spatial domain.

Unsharp Masking:

Enhances image edges while controlling the amplification of noise.

Effects:

- Enhances object edges.
- Restores fine details.
- Minimizes excessive noise amplification.

Justification:

Selectively sharpens edges rather than enhancing the entire image.

10. An application requires precise object boundary. Explain segmentation approach.

Thresholding + Canny Edge Detection

Thresholding → Separates object from background

CED → extracts object's boundaries.

Effects:

- Separates the object accurately.
- Reduces false object detection.
- Detects thin edges.

Applications:

- Shape analysis
- Object recognition
- CV Systems.